

ICT2217 Network Security
Lab Exercise 3.1: Secure Remote Access with Jump Host

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Translate the concept of Jump Host/Jump Server into real-world implementation
- Configure Virtual Routing and Forwarding (VRF) to implement Out-of-Band (OOB) management LAN
- Configure SSH for secure remote access

Required Hardware

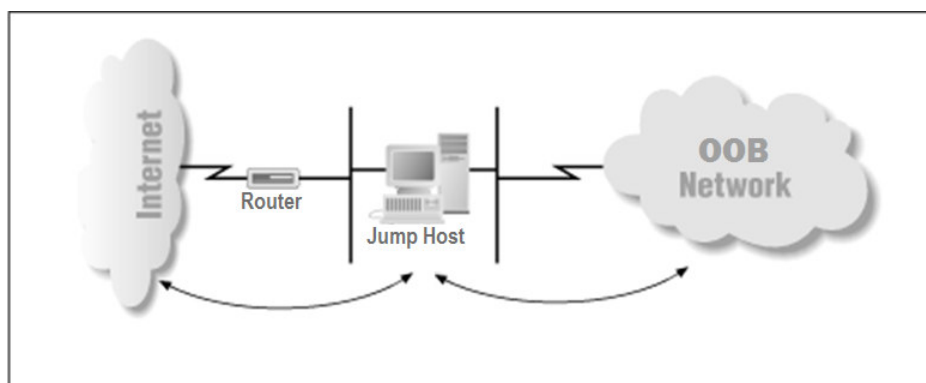
- 1 x Rack of Cisco networking devices
- 1 x Supermicro Server
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>

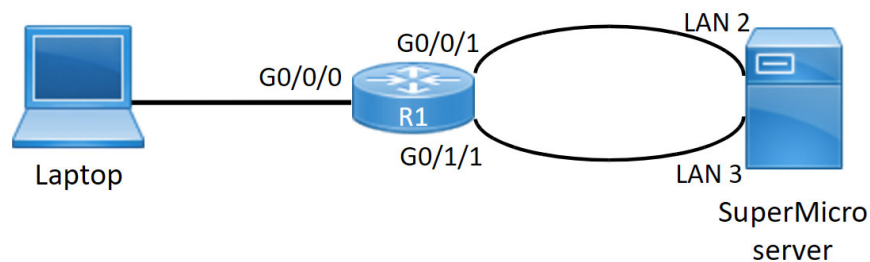
PART 1: Setting up a Simplified Network Topology for Secure Remote Access

- 1.1 For your learning benefits, this lab exercise is intentionally designed to only provide a simplified network topology but sufficient to illustrate the concept of secure remote access of network devices via a Jump Host into the Out-of-Band (OOB) network discussed in Lab3.1Notes Figure 9 as replicated below: (Your team will need to understand and extend it for your team project.)



1.2 Set up the following simplified network topology, where:

- Laptop – represent a network engineer from the Internet accessing via the edge router R1 of the enterprise network to the Jump Server hosted at the SuperMicro server;
- Server – hosting a Jump Server with one interface accessible from the Internet, and the other interface connected to the OOB network;
- R1 G0/1/1 – configured with VRF which is only accessible through the OOB network via the Jump Server.



- Refer to Appendix A Quick Start Guide to Supermicro Server, start a VM at the Supermicro server running SSH to serve as the Jump Server.
- Configure suitable IP addresses on the Laptop, R1 G0/0/0 and G0/0/1.
- Verify that the Laptop is able to ping R1 G0/0/0 and G0/0/1.
- Configure suitable IP address for the 1st interface of SSH VM linked via Linux bridge to physical port eno2 labeled as LAN 2 on the Supermicro server.
- Verify that the Laptop is also able to ping the 1st interface of SSH VM via LAN 2 successfully.

PART 2: Configuring Virtual Routing and Forwarding (VRF)

- Refer to Lab3.1 Notes on Virtual Routing and Forwarding (VRF), configure R1 G0/1/1 with suitable IP address and VRF.
- Verify that the Laptop is NOT able to ping R1 G0/1/1 directly despite able to ping G0/0/0 and G0/0/1 as verified in Step 1.5 above.
- Configure suitable IP address for the 2nd interface of SSH VM linked via Linux bridge to physical port eno5 labeled as LAN 3 on the Supermicro server.
- From R1, verify that it is able to ping VRF to the 2nd interface of SSH VM via LAN 3 using the command:

```
R1# ping vrf vrf-name VM-2nd-int-ip-addr
```

PART 3: Configuring Secure Shell (SSH) for Secure Remote Access

- 3.1 Refer to Lab3.1 Notes on Secure Shell (SSH), configure SSH on router R1.
- 3.2 You will learn more about AAA in subsequent lab exercises. For this lab exercise, configure local login to SSH as follows:
- ```
R1(config)# username usr-name secret password
R1(config)# line vty 0 15
R1(config-line)# login local
R1(config-line)# transport input ssh
```
- 3.3 Configure suitable ACL on R1 to restrict SSH access within OOB network only.
- 3.4 From the Laptop, start Tera Term or Putty to verify that it is NOT able to directly login via SSH to R1.
- 3.5 Instead, from the Laptop, login via SSH to the Jump Server hosted at the Supermicro server using IP address of the 1st interface of SSH VM configured in Step 1.6.
- 3.6 Once login to the Jump Server, are you able to ping R1 G0/1/1 from the Laptop?
- 3.7 Once login to the Jump Server, are you also able to login via SSH to R1 from the Laptop?